

Sparse-to-Dense HRTF Reconstruction

Cross-database effects of time alignment and regularised spherical-harmonic interpolation

STUDENT INFORMATION

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1. Project overview

Head-related transfer functions (HRTFs) describe how sound arriving from a direction is filtered by the listener's head and ears before reaching the two ear canals. They underpin binaural rendering, but dense measurement over a sphere is time-consuming. This project asks when a sparse directional set is sufficient for reconstruction. It will compare nearest-neighbour interpolation with unregularised and ridge-regularised spherical-harmonic (SH) reconstruction, with and without time alignment. Settings chosen on HUTUBS will be frozen and tested on SADIE II. The study will measure spectral distortion, interaural time-difference (ITD) error, interaural level-difference (ILD) error and computational cost. The analysis will be reproducible and will show whether the settings transfer across databases and where they fail. The project does not propose a new interpolation algorithm or claim universal generalisability.

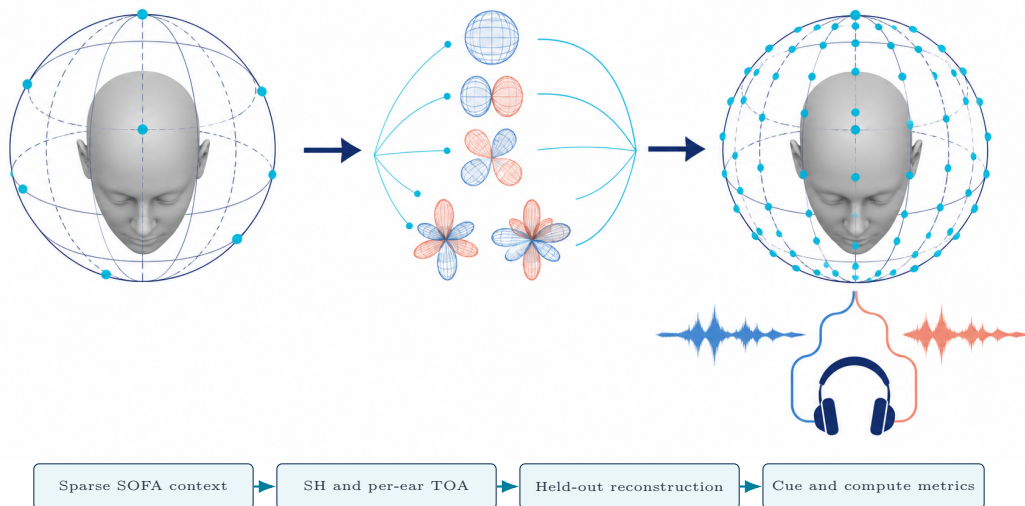


Figure 1: Conceptual workflow from sparse SOFA measurements through SH and per-ear TOA reconstruction to objective evaluation; no experimental results are shown.

2. Background and motivation

HRTF representation links several ELEC5305 topics: Fourier-domain filtering, correlation, spectral analysis, spatial sampling and binaural synthesis. Real SHs provide a compact spatial basis for HRTFs [1], yet sparse and irregular grids constrain the usable SH order [2]. Direction-dependent delay also complicates interpolation. Ear alignment and phase correction can reduce the required sampling density [3], and time alignment has been shown to change SH interpolation accuracy [4]. This topic was chosen because it turns the unit’s spatial-audio theory into a reproducible experiment that can be completed on a Mac without new measurements.

These methods are established, leaving robustness as the main question: do settings selected on one database remain useful elsewhere, and where do they fail? HUTUBS provides measured HRTFs together with associated anthropometric and geometric data [5]; SADIE II provides different subjects and sampling characteristics [6, 7]. Measurement-chain and normalisation effects create systematic cross-database differences [8]. The comparison therefore cannot be treated as a clean replication. Only pre-specified coordinate, rate and support harmonisation will be allowed. Magnitude-corrected interpolation is a stronger published method, especially in contralateral regions [9], but remains outside the core implementation. The core implementation therefore uses classical baselines and measures where they fail. Spherical neural processes offer a recent alternative [10], although neural models remain outside the present scope.

3. Proposed methodology

Measured HUTUBS and SADIE II head-related impulse responses (HRIRs) in SOFA format will be processed in Python on a Mac CPU. A versioned manifest will record URLs, licences and checksums. HUTUBS subject groups will be assigned deterministically to 60 development, 18 validation and all remaining test groups; repeat measurements stay together. Within each subject, four pre-specified farthest-point layouts at 25, 50, 100 and 200 context directions remain disjoint from target directions. All eligible SADIE II human subjects H3–H20 will be fixed before error analysis; dummy heads D1–D2 are excluded. Common preprocessing fixes coordinates, channel order, 48-kHz rate, 512-sample support and SOFA `Data.Delay` handling.

Nearest neighbour will be the reference baseline. The SH model fits complex HRTF coefficients in an orthonormal real-valued spatial basis. Order ranges from zero to the smaller of 10 and $\lfloor \sqrt{M} - 1 \rfloor$, where M is the context count. After solid-angle weighting and unit-norm SH columns, the λ grid is $\{0, 10^{-6}, 10^{-4}, 10^{-2}, 1\}$; $\lambda = 0$ isolates the effect of regularisation. Each budget and alignment condition is selected once on validation subjects, then frozen.

For alignment, left- and right-ear times of arrival (TOAs) will be estimated with a fixed 8×-upsampled, 3-kHz low-pass, 10%-of-peak onset rule after normalising `Data.Delay`. A fixed order-3, $\lambda = 10^{-4}$ SH model fitted only to context TOAs removes and later restores each ear’s delay; target TOAs are used only for final error scoring. The pre-specified fallback reconstructs minimum-phase HRIRs from interpolated log magnitude, then restores context-predicted per-ear TOA. Non-finite output or energy above ten times the input median triggers and retains the fallback. Synthetic SH round-trip error above 10^{-8} blocks experiments until repaired; if unresolved by the end of Week 7, the route stops.

Performance will be reported as solid-angle-weighted log-spectral distortion from 200 Hz to 16 kHz, absolute ITD error in microseconds, ERB-banded ILD error in decibels, fitting time, per-direction query time and model memory. ITD, ILD and spectral errors will be reported separately because these cues make distinct contributions to spatial localisation [11]. Within each database, method differences are paired at common targets but uncertainty is clustered by subject, with directions aggregated within subject. HUTUBS and SADIE II estimates remain independent. Frozen HUTUBS settings will be applied to SADIE II without retuning; success means the subject-level improvement direction over both nearest neighbour and $\lambda = 0$ remains consistent, while estimates and intervals are always reported. A fixed-direction binaural A/B WAV will be qualitative because no listening study is planned.

4. Expected outcomes

The project will deliver a working, tested Python simulation pipeline, frozen experiment configurations, machine-readable results, publication-style plots and documented commands in GitHub. Planned figures include sampling maps, reconstruction spectra, ITD and ILD distributions, directional error maps, an alignment ablation, a HUTUBS-to-SADIE II comparison and a compute-fidelity Pareto plot. Results will identify any operating region that maintains low objective cue errors in both databases and will report which directions, subjects or frequency bands fail to transfer. The final site will include the working code, methods, results, written report, a small audio demonstration and the required video report. It will not claim perceptual transparency or localisation improvement without human evidence.

5. Timeline

The schedule follows the five course phases: topic selection in Weeks 1–2, literature and data collection in Weeks 3–5, initial implementation and testing in Weeks 6–9, optimisation and evaluation in Weeks 10–11, and final reporting and GitHub documentation in Weeks 12–13. The proposal and initial GitHub Pages site are submitted in Week 4; Week 13 completes the working code, written report, final site and video report.

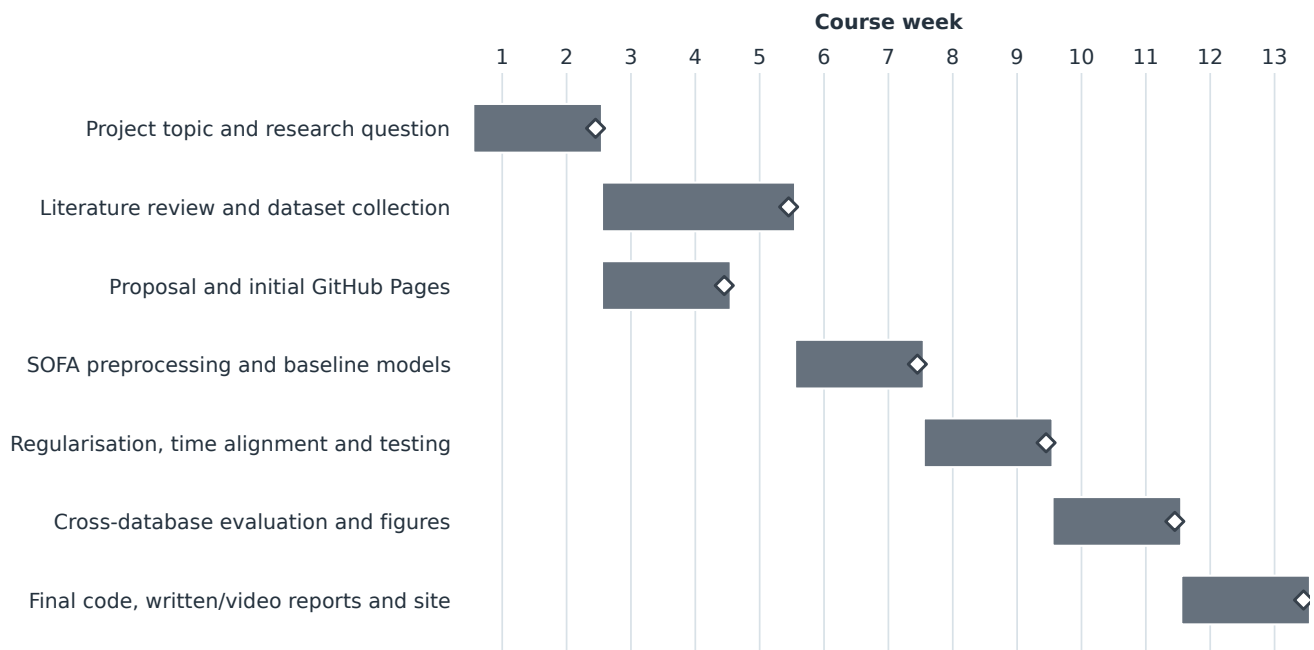


Figure 2: Course-aligned Weeks 1–13 project schedule. Grey bar position and length encode timing; diamonds mark course milestones.

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Appendix: AI use acknowledgement

AI was used in this assignment, primarily for translation and polishing. AI was also used for image generation.